

Sachin Kumar Mandal

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Professional Summary

Software Engineering fresher pursuing an MCA (2024–2026), with a strong foundational background in **C++, Data Structures, Algorithms, and Core Backend Architecture**. Eager to apply high-performance software engineering principles to design robust simulation engines, optimized algorithms, and scalable computational pipelines. Passionate about tackling complex system challenges, low-level problem solving, and building high-efficiency software solutions.

Work Experience

Full Stack Web Development Intern

Sep 2023 – Oct 2023

The Website Makers, India

- Collaborated actively with Agile teams to design, develop, and deploy robust RESTful JSON APIs via Docker containers on a Linux-based cloud infrastructure, utilizing Git for version control and consistently maintaining a 99.9% system uptime.
- Engineered backend integrations utilizing Node.js and Python scripting to automate data validation, enforce web security protocols, and conduct rigorous API testing via Postman.
- Executed highly optimized SQL scripts within PostgreSQL to facilitate rapid backend data extraction; actively evaluated and resolved bug submissions, improving average query response times by 35%.
- Authored comprehensive technical documentation for API endpoints and web services, fostering cross-functional alignment and strict adherence to enterprise software delivery standards.

Projects

High-Performance Algorithmic Data Engine

C++, STL, Data Structures, Algorithms

- Designed and implemented modular, memory-efficient C++ data structures and custom algorithms utilizing STL containers to process and manipulate complex data streams with optimized runtime complexity.
- Focused on rigorous logical structuring, memory management, and performance optimization to handle large-scale data processing requirements efficiently.

AI-Augmented Social Media Platform

Python, PyTorch, LangChain, RAG, LLMs

- Developed an AI-integrated backend utilizing Python, PyTorch, and LangChain to process complex data workflows and connect with Large Language Models (LLMs) like OpenAI/DeepSeek.
- Implemented Retrieval-Augmented Generation (RAG) techniques to dramatically improve the accuracy of AI-generated responses and information extraction for user-facing features.

Leave Management System

C++, Node.js, PostgreSQL, Jenkins, GitLab CI

- Architected scalable backend components and core business logic engines across the SDLC; implemented rigorous computational workflows and secure data exchange routines.
- Integrated automated Jenkins and GitLab CI/CD pipelines within a robust UNIX/Linux server environment to streamline software deployment, cutting release cycle time by 40%.
- Resolved critical database caching bottlenecks and optimized data retrieval paths, boosting PostgreSQL data execution speeds by 45% to ensure optimal system performance.

Education

Radha Govind University, Jharkhand

2024 – 2026

Master of Computer Applications (MCA) – Computer Science STEM

Dr. Shyama Prasad Mukherjee University

Aug 2021 – Jul 2024

Bachelor of Science (B.Sc.) Information Technology – Aggregate 60%+, No Arrears

Skills

Programming & Core: C++, C, Python, JavaScript, TypeScript, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), STL

Digital Design & Scripting: Verilog/VHDL (Basic), Tcl, Python Scripting, Shell Scripting (Bash/Awk)

Cloud, OS & DevOps: Git, CI/CD (Jenkins, GitLab CI), Docker, Linux/UNIX Environments

Databases & Tools: PostgreSQL, SQL, REST APIs, Postman, Agile/Scrum, SDLC, Unit Testing

Certifications & Achievements

- **Superset Top Performer:** Scored 10/10 in strict Live Coding software engineering assessments.
- **JPMorgan Chase & Co. (2025 – 2026):** Software Engineering Virtual Experience with exposure to core engineering practices.
- **SAP Certified:** SAP Build Apps & Fiori UX Design – enterprise software architecture fundamentals.
- **Deloitte Australia (2026):** Authored technical documentation for regulatory standards and system compliance.